European Commission

CAMSS – Solutions

Core Standards and Specifications Vocabulary (CSSV)

V2.0.0

Specification

Directorate-General for Digital Services

interoperable europe

Reference
Core Standards and Specifications Vocabulary v2.0.0

Keywords
<CAMSS, CAV, CSSV, assessment, Controlled vocabularies, EIF>

CHANGE CONTROL

Modification	Details
Version 2.0.0	
Current version	**This is major release of CSSV includes the following changes:** **- Best practices on namespace definition in the CSSV T-Box.** **- Addition of CSSV SHACL, containing constraints from CSSV T-Box v1.4.0.** **- Simplification of OWL constructs and external dependencies.** **- Refined modelling of cssv:ApplicationProfile, cssv:Standard and prof:Profile.**
Version 1.4.1	
Current version	**This is a bug fixing release of the CSSV Vocabulary including the following changes:** **- Best practices on namespace definition in the CSSV T-Box: empty prefix and base were removed.** **- OWL representations of external terms redefining their meaning have been removed from the T-Box.** **- References to the redefinition of these late terms have been removed from this document for consistency.** **- A restriction on the dcat:previousVersion has been added with regards to the CSSV Specification class.**
Version 1.4.0	
Last version	**Main changes for this minor version of the Vocabulary are:** **- Alignment with SEMIC style guide practices.** **- CSSV update regarding with new versions of DCAT v3.** **- In the T-box, a new property for recommending the use of specifications was added.**
Version 1.3.0	
	Main changes for this minor version of the Vocabulary are: - Refinement of Specification and Standard's definition. - Refinement of Specification's versioning. - Alignment of CSSV T-box and CSSV specification. - Adoption of the schema:contactPoint relationship for expressing the Agent who maintains the Specification. - Recovery of the cssv:alternative attribute. - Addition of the dcat:downloadURL attribute.
Version 1.2.0	

	This minor release of the CSSV includes: - The class cssv:Scope has been removed. - The cssv:Specification class inherits from the DCAT the attribute dcat:Version to identify the version of the specification. - Addition of the attribute cssv:acronym in the class cssv:Specification and the attribute dct:alternative has been removed.
Version 1.1.0	
	This minor release of the CSSV includes: - Use of the latest of version of DCAT. - The class adms:Asset has been replaced by dcat:Resource. - The dcat:Resource class has a dct:creator which is a foaf:Agent. - Addition of the class schema:ContactPoint.
Version 1.0.0	
Initial version	

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Foreword

This Specification has been produced by the Common Assessment Method for Standards and Specifications (CAMSS) Team, a Digital Europe (DEP) Programme initiative in alignment with the European Standardisation Regulation 1025/2012.

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Introduction

1. Introduction

The CSSV is the vocabulary used for the information exchange related to standards and specifications amongst software solutions, as well, it is the key element for the development and maintenance of the EIRA Library of Interoperability Specifications (ELIS).

1.1 Context

The DEP Programme of the European Commission supports the development of solutions that enable the cross-border delivery of interoperable public services in Europe. In order to ensure the interoperability of those services, the EIA action works as an integrator between the Member States and other departments of the European Commission for the development of a joint interoperability architecture for public services. The main output of this action is the European Interoperability Reference Architecture (EIRA©).

As an element of EIRA©, the EIRA Library of Interoperability Specifications (ELIS) was created. The ELIS contains the specifications describing the interoperability requirements of the architecture building blocks (ABBs) that conform to EIRA©.

At the core of the ELIS, there is also another asset developed in the context of the CAMSS action that shall be referenced which has been further developed: the Core Standards and Specifications Vocabulary (CSSV).

1.2. Objective and Scope of the document

The objective of this document is to provide an interoperability-oriented solution for the information exchange related to standards and specifications amongst software solutions, the Core Standards and Specifications Vocabulary.

The scope of this document encompasses the following.

- Conceptual data models used for the CAMSS Vocabulary.
- Constraints and rules specific to the CAMSS domain.
- A reference implementation of the A-Box as an OWL and SHAPES Turtle syntax (Annex 1).

In addition, this vocabulary has been publicly analysed to create a stable version of the vocabulary.

The CSSV has been reviewed by a group of experts contributing to the new release of the vocabulary.

1.3 Methodological approach

The CSSV defined in this document is a major release of the vocabulary. As its precedent release, the CSSV is created for the development of the ELIS.

The approach followed for the development of the CSSV adheres to three fundamental principles.

1. Reuse and share when possible (i.e., do not reinvent the wheel).
2. Do not betray the knowledge and experience of the domain, nor the terminology and interpretation of the concepts (i.e., do not invent new terms when they already exist in the communities of practice or generic domains).
3. Isolate technical and business constraints and rules as much as possible (i.e., externalise them in separate artefacts, for example, graph and data shapes for the control and validation of the data). This has a great impact on the quality and cost of the implementation, as well as the maintenance of the vocabulary.
4. One way of facilitating the semantic interoperability consists of reusing existing generic ontologies and vocabularies. This way, the semantics of common concepts and properties are agreed upon without the need for re-discussion. When concepts or properties have not been identified nor defined for the purposes pursued, they must be proposed as either extensions, or from scratch.

The methodological approach followed for the development of the CSSV reuses the following ontologies and vocabularies.

- Data Catalogue Vocabulary (DCAT);
- Friend of a Friend (FOAF);
- The W3C Organization Ontology;
- DCMI Metadata Terms (DCTerms);
- The Profiles Vocabulary (PROF);
- Schema.org.

The rationale for defining this vocabulary goes as follows.

1. In the context of the European Commission, the categorisation of standards and specifications based on their purpose, scope and origin are paramount for the their implementation as interoperable solutions.
2. Existing concepts in other ontologies did not cover all the information requirements needed in CSSV and had to be reused or specialised by new classes.
3. Concepts and properties existing in other ontologies have different semantics to those needed in CSSV.
4. Concepts required in CSSV have not been identified in any other existing ontologies and therefore needed to be defined.
5. Given this is “Core” vocabulary, a key goal is to make it as flexible as possible. This means that predicates are set with optional and multiple cardinality (0..n) unless there is a strong reason for further restriction.

1.4 Structure of this document

This document consists of the following sections.

- Section 2 describes the related solutions to the Core Standards and Specifications Vocabulary (CSSV).
- Section 3 explains the CSSV model and identifies the classes and properties defined for the vocabulary.
- Section 4 contains the Conformance Statement for this vocabulary.
- Section 5 describes how CSSV is compliant with the FAIR principles.
- Section 6 describes specific accessibility and multilingualism aspects.
- Section 7 lists the different acronyms used in the whole document.
- Section 8 contains related references.

2

RELATED SOLUTIONS

2. Related Solutions

This section lists the different CSSV related solutions. Note that some are still under development.

2.1 EIRA Library of Interoperability Specifications (ELIS)

The ELIS is a family of interoperability specifications that define the interoperability aspects of the Architecture Building Blocks (ABBs) contained in EIRA©. ELIS aims to support architects for the modelling of solutions based on EIRA©. The current version of ELIS will need to be slightly revamped to accommodate the concepts defined in the CSSV and to support the requirement of all stakeholders, e.g., EIRA-based solution developer needs, NATO profiles, and others.

2.2 Core Assessment Vocabulary (CAV)

The Core Assessment Vocabulary represents, expresses, and defines what an “Assessment” of “Assets” is and how to perform the assessment based on “Criteria”. It is a domain-agnostic vocabulary, meaning that it can be used to assess any asset.

2.3 Data Catalogue Vocabulary (DCAT)

The Data Catalogue Vocabulary (DCAT) is used to describe public sector datasets in Europe. This vocabulary has been developed by the W3C. DCAT can be used to describe any type of asset (treated as a dataset, especially when considering that metadata is also data).

The figure below shows the DCAT conceptual data model with its classes and properties:

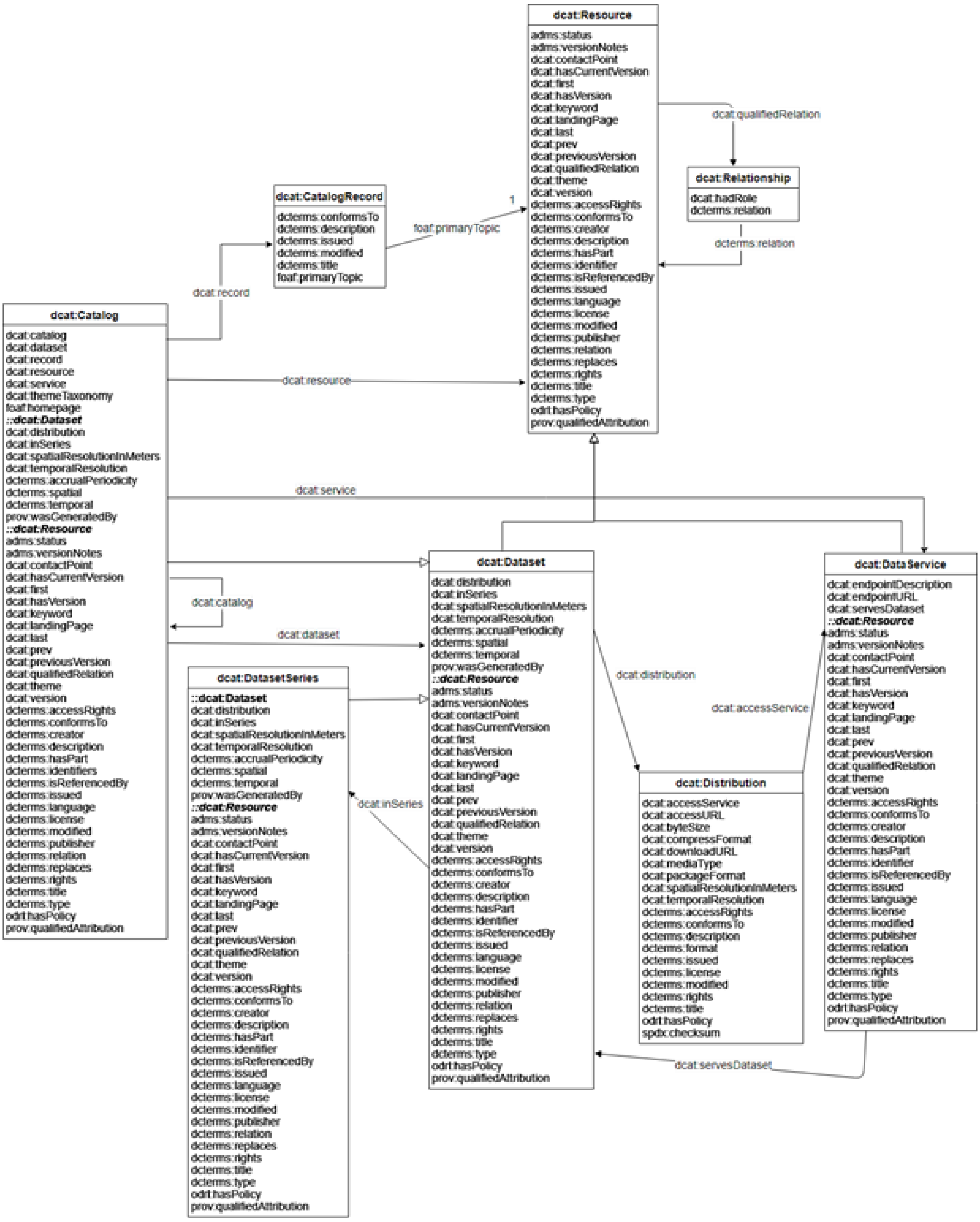

Figure 1: DCAT classes and properties

In the CSSV model, the class Specification can be considered the “root” class and is a “Resource” as defined in DCAT.

3

CORE Standards and Specifications VOCABULARY (CSSV)

3. Core Standards and Specifications Vocabulary (CSSV)

The CSSV is the vocabulary used for the information exchange related to standards and specifications amongst software solutions, as well, it is the key element for the development and maintenance of the EIRA Library of Interoperability Specifications (ELIS).

The Core Standards and Specifications Vocabulary is depicted in Figure 2 CSSV Data model. The figure shows the classes and properties that are used or defined in the CSSV.

3.1 Data Model for the CSSV

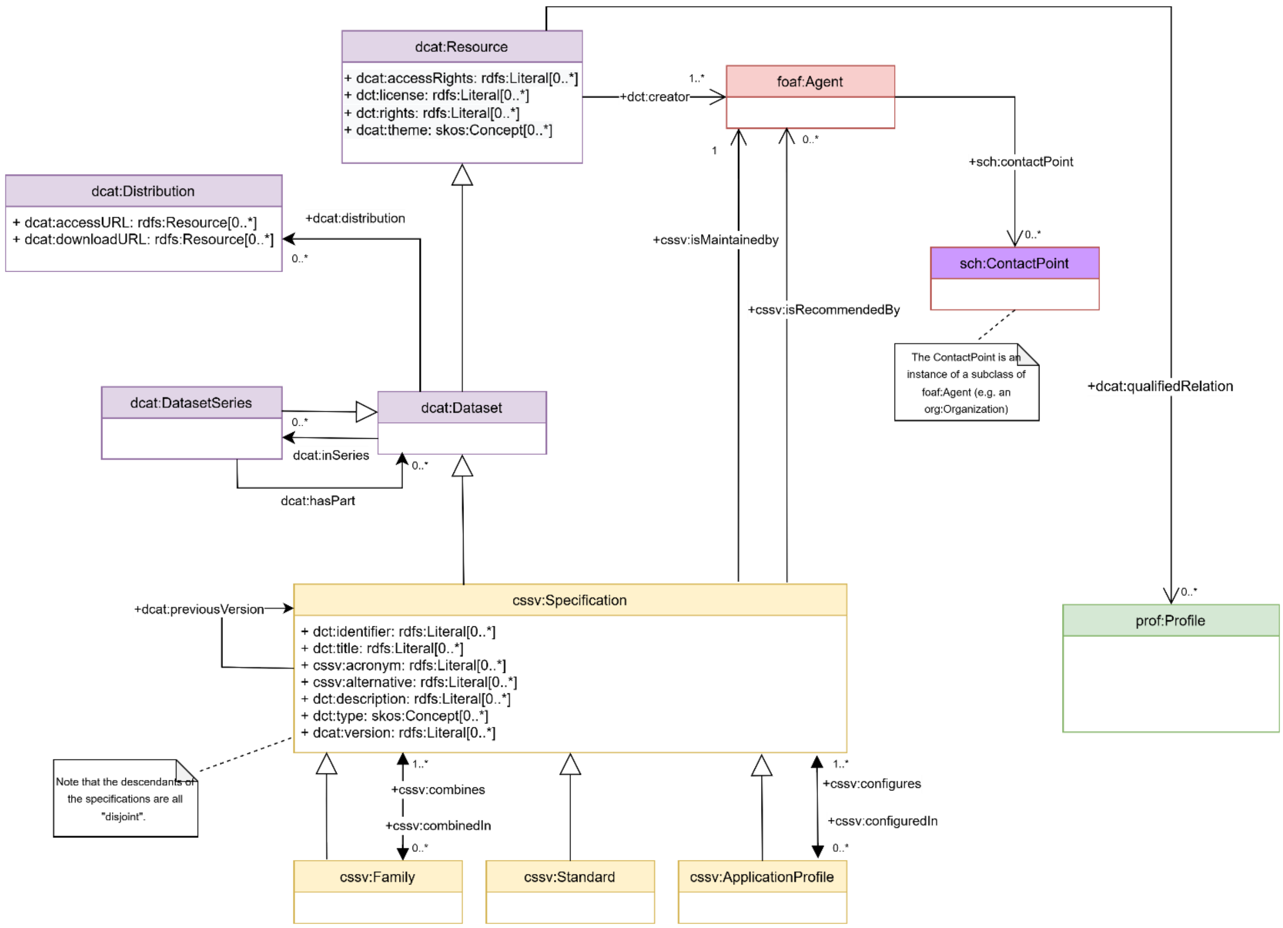

Figure 2: The Core Standards and Specifications Vocabulary

3.1.1. *Structuring CSSV through external vocabularies*

Regarding the scope of cssv:Specification, a new paradigm shift is essential moving beyond its original framing as an adms:Asset to embrace a broader interpretation that extends its initial role as a semantic asset. Outside the CSSV context, multiple sources -i.e., vocabularies, institutions, guidelines, offer well-established definitions of standards and specifications typically grounded on a descriptive and agnostic contexts. Reviewing these sources has informed the approach taken in this major version of the CSSV.

- Dublin Core Terms (DCT): which provides general metadata classes and properties for expressing conformance relationships between a resource and a standard.
- Profiles Vocabulary (PROF): aimed at increasing interoperability by defining constraints, extensions, or combinations of standards and specifications.
- SEMIC Style Guidelines: which provide best practices for modeling and documenting conceptual models, ontologies, data specifications, and data application profiles.
- Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs (DG GROW): which coordinates and supervises the EU standardisation system from a Single Market policy perspective and incorporates European standards (EN) and harmonised European standards (hENs).
- The Interoperable Europe Act (IEA): which establishes a formal definition of interoperable solutions, encompassing relevant specifications and standards.

While DCT and DCAT underpin the CSSV, the European standardisation context demands a more rigorous interpretation of what a standard and a specification constitutes, with the ability to extend seamlessly to international frameworks.

Starting from the base of the foundation, DCT defined a dct:Standard class as a reference point that determines how well a thing performs or how it differs from alternatives. Eventually, DCT has helped to describe specifications and their relationship in the PROF, articulating the prof:Profile and dct:Standard classes and properties like dct:title, dct:description and dct:relation. This set a precedent for CSSV 1.0.0 to adopt a similar path in 2020 declaring a cssv:Specification as a dcat:Dataset. Over time, the grounded need to express a specification from a European position has led to adopt a more comprehensive understanding of what a specification implies, which has questioned the reliability of the current CSSV structure, notably with regards to dct:Standard and prof:Profile.

Following the emergence of bodies and regulations that govern specifications, a standard and a specification become concise entities which differ from each other on the manner they are described and the dimensions where they are adopted. In this sense, it was logical to maintain the cssv:Specification and the cssv:Standard distinction; moreover, the disjoint character of the cssv:Standard with regards to cssv:ApplicationProfile and cssv:Family, and the other way around enable to:

- (use case 1.1) Distinguish when a specification defines constraints as a cssv:ApplicationProfile, independently of its derivation from an individual which is part of a family of specifications (cssv:Family);
- (use case 1.2) Distinguish when a specification defines constrains as a cssv:ApplicationProfile, independently of its derivation from an well-established and widely adopted specification (cssv:Standard);
- (use case 2.1) Redefine a specification which attained a wide adoption as a cssv:Standard and recreate as a cssv:ApplicationProfile;
- (use case 2.2) Redefine a specification which attained a wide adoption as a cssv:Standard and recreate in combination with related specifications as a cssv:Family

- (use case 3.1) Redefine a part of a family of specifications and recreate into a cssv:ApplicationProfile potentially combining with other parts of the precedent family of specifications;
- (use case 3.2) Redefine a part of a family of specifications and recreate into a cssv:Standard either combining with other parts of the precedent family of specifications or behaving as an independent specification.

There is a clear agentive perspective which is not thoughtfully tackled in other vocabularies, where only a descriptive perspective exists. As the DCT vocabulary states, the dct:Standard is a “reference point against which other things can be evaluated or compared” and through dct:conformsTo and dct:format “things” can be evaluated or compared on a compliance basis (e.g., compliance to benchmarks, specific considerations, rules) or a representative basis (e.g., formats). This articulation goes further on the PROF which reuses dct:relation to dip on the specialisation and description of these specialised “base specifications”.

Finally, another important matter regarding the evolution of CSSV, is the relation of the classes cssv:ApplicationProfile and cssv:Standard with regards to prof:Profile and dct:Standard, respectively. Currently, the relation between cssv:ApplicationProfile and prof:Profile, while they sometimes can have some conceptual overlapping, there is no need to create a direct relation between them: PROF does not limit to define constraints in base specifications but also extends a base specification in a general basis. Moreover, the cssv:Standard class is currently related as a dct:Standard, that allows to the CSSV to inherit the definition already stablished by DCT. However, this relation may be misleading since dct:Standard does not introduce any extensions beyond the general concept DCT provides (even in DCAT a resource’s conformance can be performed on a comparison basis with regards to a dct:Standard).

3.1.2. Interpretation

The main class of the CSSV model is the “Specification”. As represented in the conceptual model of the CSSV, a Specification is an asset, since it inherits from the dcat:Dataset, which inherits from the dcat:Resource, defined as ‘a collection of data published or curated by a single agent and made available in one or more representations.’. A Specification can serve to multiple function, not merely limited to semantic assets (as ADMS defines), but also describes requirements; and these requirements, depending on their purpose and context of use, can converge into a Standard, an Application Profile, a Family or a collection of other specifications.

The CSSV model defines the Specification and three additional specialisations:

- A Specification, which describes precise requirements that are needed for the implementation of a solution. A specification is not necessarily a standard.

- A Standard as a Specification that has reached a certain maturity and widespread adoption and potentially endorsed, meaning that is recognised and supported by a community or sanctioned by an authority. Normally, a Standard is a Specification that is part of a interoperable solution implementation.

- An Application Profile as customisation of one or more existing specifications potentially for a given use case or a policy domain adding an end-to-end narrative describing and ensuring the interoperability of its underlying specification(s). By customisation, we understand the “addition of more specificity by identifying mandatory, recommended, and optional elements, as well as by defining controlled vocabularies to be employed”.

- A Family as a collection of interrelated and/or complementary specifications, standards or application profiles and the explanation of how they are combined, used, or both.

Optionally, the use of the dct:type property in cssv:Specification indicates the "type" of a specification, whether it represents a data specification, data profile, data artifact, or reference model etc. The dct:type property, as defined in the DCT vocabulary, may serve to complement and classify a specification according to its particular kind, function, or purpose. There exist referenced codelists provided by ADMS-AP and the Publications Office of the EU (PO) that may be reused.

When one of the specialisations of Specification does not adapt to a specific context and needs to work in conjunction with a other Specifications, the use of a collection of Specifications is highly recommended through the dcat:DatasetSeries. The dcat:DatasetSeries class represents a collection of datasets that are published separately, but share some characteristics that group them. The specialisation of a Specification are linked to other Specifications by using the property dcat:inSeries (being dcat:hasPart the inverse relation to it). Note that a collection of Specifications can also be hierarchical, and such collection can be a member of another collection of Specifications.

The use of dcat:DatasetSeries is not always limited to representing a specialisation of a Specification, it can also be applied to a Specification. Unlike a Family of Specifications, the Specifications within a collection are grouped without requiring an explicit relationship among them, allowing for a more flexible and loosely coupled collection. There are occasions where collections of Specifications are applied to a context or a domain in a specific "configuration". Thus application profiles may conform sets of "themed" specifications. For this, the CSSV model uses the property "configures/includedIn" and the dcat:theme property pointing at a skos:Concept (i.e. a code, see the DCAT model above). It is important to note that the descendants of the specifications are all "disjoint". Thus, Application Profiles and Families are Specifications that refer to or put together with other Specifications and/or Standards, but cannot themselves be considered Standards. This entails that an individual of an application profile or family cannot be a standard, but does not preclude that, in time, the application profile or the family can become standards. If that were the case then individuals of cssv:Standard would be created to represent the standardisation of those specifications that are Application Profiles and Families.

Another aspect to consider is the evolution of a Specification. One Specification, in time, may become a Standard. In these cases, the authority (author) that defined the Specification may be different from the one that creates and maintains artefacts out of the Standard. Think, for example, of the artefacts produced, maintained and distributed by the Publications Office of the EU (OP) in its site EU Vocabularies: all these artefacts are defined by other authorities (e.g. the ISO), whilst the artefacts (e.g. the controlled vocabularies expressed in SKOS, XML, GeneriCode, XML, etc.) are supplied by the OP. For this, the CSSV uses the properties dct:creator and cssv:isMaintainedBy; where dct:creator (as for dct:publisher) DCT associates a URL as property range.

The maintainer of a Specification is a foaf:Agent, which allows great flexibility to the CSSV model as foaf:Agent is the base class in many ontologies. Moreover, a foaf:Agent may represent a member state that recommends the a Specification; the property cssv:isRecommendedBy is used for this end. The CSSV puts forward the reuse of the Core Person Vocabulary (CPV) and the Organization Ontology (W3C Org) for this purpose. Also, the foaf:Agent also provides the contact point of the specification.

Concerning the intellectual property rights (IPR), they are covered by the fact that a Specification which is a dcat:Resource and it allows to define the dct:license and dct:rights. Further extensions of licencing matters and derivation of IPR can be modelled using PROV-O.

A new shift to CSSV 2.0 is the discontinuation of cssv:ApplicationProfile as a subclass or direct specialisation of prof:Profile. While this interpretation is technically compatible with the definition provided in the PROF specification, where a profile is understood as, "a specification that constrains, extends, combines, or provides guidance or explanations about the usage of other specifications", it surpasses the broader field of application foreseen for the CSSV. To overcome this, the CSSV introduces a more flexible and semantically neutral mechanism based on dcat:qualifiedRelation, allowing relationships between Specifications and a prof:Profile. This approach replaces the previous direct link between prof:Profile and cssv:ApplicationProfile, ensuring that the connection remains optional and context-driven rather than structural. This will avoid to not enforce to many constrains and links that are not semantically necessary. Since ideally CSSV wants to relate concepts that are similar to each other without enforcing how things are used, nor creating unnecessary restrictions for existing ontologies. To ensure clarity, the use of dct:relation within each dcat:qualifiedRelation instance is made mandatory.

In summary, this new structure ensures conceptual clarity and traceability within the CSSV model, allowing consistent classification and evolution of specifications across different levels of maturity and reuse.

3.2 Class:Specification

OWL Class	cssv:Specification
Label	Specification
Definition	Document with normative guidelines or characteristics that can be used consistently to ensure that materials, products, processes and services conform to one or a set of requirements (ISO). Specifications are the cornerstone of interoperability, as they set the grounds for the design and development of interoperable and reusable solutions based in common services at the lowest level.
Subclass of	dcat:Dataset

3.2.1 Property: previousVersion

OWL Property	dcat:previousVersion
OWL type	owl:ObjectProperty
Label	previousVersion
Definition	The previous version of a resource in a lineage.
Usage note:	The URI identifying the previous version of the specification.
Cardinality	0..n

3.2.1 Property: configuredIn

OWL Property	cssv:configuredIn
OWL type	owl:ObjectProperty
Label	configuredIn
Definition	A set of Specifications potentially for a given use case or policy domain that are aggregated in an ApplicationProfile.
Domain	cssv:Specification
Range	cssv:ApplicationProfile
Cardinality	0..n

3.2.1 Property: combinedIn

OWL Property	cssv:combinedIn
OWL type	owl:ObjectProperty
Label	combinedIn
Definition	A set of Specifications that are complementary and interrelated, forming a Family of Specifications.
Domain	cssv:Specification
Range	cssv:Family
Cardinality	0..n

3.2.1 Property: isMaintainedBy

OWL Property	dct:type
OWL type	owl:ObjectProperty
Label	isMaintainedBy
Definition	The Person, Organisation responsible to update and maintain the specification.
Domain	cssv:Specification
Range	foaf:Agent
Cardinality	1..n

3.2.1 Property: *isRecommendedBy*

OWL Property	dct:type
OWL type	owl:ObjectProperty
Label	isRecommendedBy
Definition	A link to any other person or organisation that recommends the specification.
Domain	cssv:Specification
Range	foaf:Agent
Cardinality	0..n

3.2.1 Property: *identifier*

OWL Property	dct:identifier
RDF type	rdf:Property
Label	identifier
Definition	An unambiguous reference to the resource within a given context.
Range	rdfs:Literal
Property Type	xsd:anyURI
Usage note:	This property contains the main identifier for the specification, e.g. the URI or another unique identifier.
Cardinality	0..n

3.2.1 Property: title

OWL Property	dct:title
RDF type	rdf:Property
Label	title
Definition	A name given to the resource.
Range	rdfs:Literal
Property Type	xsd:string, rdf:langString
Usage note:	The name given to the specification.
Cardinality	0..n

3.2.1 Property: acronym

OWL Property	cssv:acronym
OWL type	owl:DatatypeProperty
Label	acronym
Definition	Abbreviation of the specification.
Range	rdfs:Literal
Property Type	xsd:string, rdf:langString
Cardinality	1

3.2.1 Property: alternative

OWL Property	**cssv:alternative**
OWL type	owl:DatatypeProperty
Label	alternative
Definition	The alternative name of the specification. Additional information: The alternative title for a Specification is optional. CAMSS aims to standardise the names of specifications as a reference at EU level. However, the title of a Specification might, distinct to the provided by CAMSS, may be widely used in the community.
Range	rdfs:Literal
Property Type	xsd:string, rdf:langString
Cardinality	0..n

3.2.1 Property: description

OWL Property	**dct:description**
RDF type	rdf:Property
Label	description
Definition	An account of the resource.
Range	rdfs:Literal
Property Type	xsd:string, rdf:langString
Usage note:	This property contains a free-text account of the specification. This property can be repeated for parallel language versions of the description.
Cardinality	0..n

3.2.1 Property: type

OWL Property	dct:type
RDF type	rdf:Property
Label	type
Definition	The nature or genre of the resource.
Usage note:	This property refers to the type of the specification. A controlled vocabulary for the values has not been defined for the time being.
Cardinality	0..n

3.2.1 Property: accesRights

OWL Property	dct:accessRights
RDF type	rdf:Property
Label	accessRights
Definition	Information about who can access the resource or an indication of its security status.
Cardinality	0..n

3.2.1 Property: license

OWL Property	dct:license
RDF type	rdf:Property
Label	license
Definition	A legal document giving official permission to do something with the resource.
Usage note:	A legal document under which the resource is made available.
Cardinality	0..n

3.2.1 Property: rights

OWL Property	dct:rights
RDF type	rdf:Property
Label	rights
Definition	Information about rights held in and over the resource.
Usage note:	A statement that concerns all rights not addressed with dct:license or dct:accessRights, such as copyright statements.
Cardinality	0..n

3.2.1 Property: version

OWL Property	dcat:version
OWL type	owl:DatatypeProperty
Label	version
Definition	The version indicator (name or identifier) of a resource.
Usage note:	The version number of a resource. This is a free-text string, typical values are “1.5” or “21”. The URI identifying the previous version can be provided using dcat:previousVersion.
Cardinality	0..n

3.3 Class: Standard

OWL Class	cssv:Standard
Label	Standard
Definition	Specification that has reached a certain maturity and widespread adoption and potentially endorsed.
Subclass of	cssv:Specification, dct:Standard

3.4 Class: ApplicationProfile

OWL Class	cssv:ApplicationProfile
Label	ApplicationProfile
Definition	An application profile “customises one or more existing specifications potentially for a given use case or a policy domain adding an end to end narrative describing and ensuring the interoperability of its underlying specification(s)”.
Subclass of	cssv:Specification

3.4.1. Property: configures

OWL Property	cssv:configures
OWL type	owl:ObjectProperty
Label	configures
Definition	Whether an Application Profile design or adapts a Specification for a specific purpose.
Domain	*cssv:ApplicationProfile*
Range	*cssv:Specification*
Cardinality	0..n

3.5 Class: Family

OWL Class	cssv:Family
Label	Family
Definition	A family is a collection of interrelated and/or complementary specifications, standards, or application profiles and the explanation of how they are combined, used, or both.
Subclass of	cssv:Specification

3.4.3. Property: combines

OWL Property	**cssv:combines**
OWL type	owl:ObjectProperty
Label	combines
Definition	Whether a Family is a union of more than one Specifications.
Domain	*cssv:Family*
Range	*cssv:Specification*
Cardinality	0..n

4

CONFORMANCE STATEMENT

4. Conformance Statement

A data interchange of Standards or Specifications, however that interchange occurs, is conformant with the CSSV if:

- it uses the terms (classes and properties) in a way consistent with their semantics as declared in this specification;
- it does not use terms from other vocabularies instead of ones defined in this vocabulary that could reasonably be used.

A conforming data interchange:

- may include terms from other vocabularies;
- may use only a subset of CSSV terms.

The CSSV is technology-neutral and a publisher may use any of the terms defined in this document encoded in any technology although RDF and XML are preferred.

5

FAIR principles conformance

5. FAIR principles conformance

The CSSV is compliant with the following aspects of the FAIR (Findable, Accessible, Interoperable, Reusable) principles:

Findable:

- The main properties of the CSSV have a unique identifier throughout it and the metadata is registered with the identifier as the description. The properties are also indexed through their classes.

e.g.

Each class in the CSSV, such as cssv:Specification, can be assigned with a unique identifier. Metadata associated with cssv:Specification is stored and indexed, making it searchable and easy to locate by its identifier.

Accessible:

- The CSSV is an open source element, meaning that is free, open and universally implementable.

e.g.

The CSSV is hosted in a public GitHub repository and is accessible via an open URL (e.g., https://github.com/isa-camss/CSSV). Anyone can download, implement, or contribute to the vocabulary without restrictions.

Interoperable:

- The CSSV is based on open specifications. Furthermore the data use a formal, accessible, shared, and broadly applicable language for knowledge representation.

e.g.

The CSSV defines terms using RDF (Resource Description Framework), a standard, machine-readable format, ensuring that data can be seamlessly exchanged between systems regardless of their underlying platforms.

Reusable:

- In the CSSV the data is structured so it can be used in multiple settings, in this sense it is a domain agnostic vocabulary.
- The CSSV can be extended for designing new data models according to the users' needs, while still ensuring the interoperability.

e.g.

The property cssv:identifier is defined generically, allowing it to be used for any type of entity (e.g., persons, organisations, or products). This flexibility ensures that the CSSV can support data models in various domains.

6

ACCESSIBILITY AND MULTILINGUAL ASPECTS

6. Accessibility and Multilingual Aspects

The CSSV can operate in any language due to the following.

- In a multilingual context, with all properties that are datatype “Text”, the value may exist in multiple languages, the property may be instantiated multiple times and tagged with the language identifier for the value used for that property.
- The CSSV specification encourages the use of PURIs as identifiers.

7

ACRONYMS

7. Acronyms

Term	Description
CAV	Core Assessment Vocabulary
EIRA©	European Interoperability Reference Architecture
EU	European Union
CSSV	Core Standards and Specifications Vocabulary
DCAT	Data Catalogue Vocabulary
ELIS	EIRA Library of Interoperability Specifications
ABBs	Architecture Building Blocks
SEMIC	Semantic Interoperability Community
PAV	Provenance, Authoring and Versioning
PROF	The Profiles Vocabulary
SKOS	Simple Knowledge Organization System
RDF	Resource Description Framework

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Annex 1

CSSV Model

Annex 1: CSSV Model

CSSV_UML_v2.0.0.drawio

CSSV Tbox

CSSV Shapes